VYOMA — The Offline AI Receptionist

Expected Judges' Questions and Suggested Answers

Science Fest Preparation Guide | Class 8 Project Team

How to use this guide:

- Read every question aloud and practise answering in your own words — do not memorise word-for-word.
- Judges love it when you say "I don't know, but here is my guess..." more than a wrong confident answer.
- Decide in advance which teammate answers which topic, but both of you should be able to answer the basic questions.
- Keep answers short (20–30 seconds). If the judge wants more, they will ask.

Your 30-Second Opening Pitch (practise this!)

"Vyoma is a talking receptionist robot that works completely offline — no internet needed. A visitor walks up, the camera notices them, they ask a question by speaking, and Vyoma listens, thinks using a small AI brain running on this laptop, and answers aloud. It answers only from our own documents and school website, so it never makes up outside information. It can also note down who came to visit and inform the staff member. All visitor data stays on this machine — nothing goes to the cloud."

A. Basic Questions (almost guaranteed)

Q1. What does your project do? Explain in one minute.

Answer: Vyoma is an AI receptionist. When a visitor speaks, a microphone records the voice, a speech-to-text program converts it into words, an AI language model reads those words along with facts from our knowledge base and decides the reply, and a text-to-speech program speaks the answer aloud. It can also detect a person with the camera, log the visit in a database, and send a notification to staff. Everything runs on one ordinary laptop with no internet.

Q2. Why did you name it Vyoma?

Answer: Vyoma is a Sanskrit word meaning 'sky' or 'space'. We liked it because it sounds like a friendly Indian name for an assistant, and it is also the name ISRO gave to its humanoid robot concept (Vyommitra), which inspired us.

Q3. What made you choose this project?

Answer: Every office and school has a reception desk, and visitors often wait when the receptionist is busy. We wanted to see if a computer could greet visitors and answer common questions by itself. Also, most AI assistants like Alexa need the internet — we wanted to prove a useful AI can run fully offline.

Tip: Add your own personal story here if you have one — judges like genuine motivation.

Q4. Is this hardware or software? What did you build yourselves?

Answer: It is mainly a software project running on a laptop with a webcam and microphone. We did not train the AI models ourselves — that needs huge computers. What we built is the system that connects them: the listening loop, the knowledge base of our organisation's information, the visitor database, the tools the AI can use, and the configuration that makes it behave like a polite receptionist.

Tip: Be honest about this — judges respect students who clearly separate 'what we used' from 'what we made'.

B. How It Works (technical questions)

Q5. Explain the steps from when I speak to when the robot replies.

Answer: Five steps. (1) The microphone records until it detects silence. (2) A program called Whisper converts the speech into text. (3) The question is compared with our knowledge files, and the most similar facts are picked out. (4) A language model called Qwen reads the question plus those facts and writes a short reply. (5) A program called Piper converts that reply into a voice and plays it through the speaker. The whole cycle takes a few seconds.

Q6. What is an LLM / language model? How does it 'think'?

Answer: A large language model is a program trained on a huge amount of text. It learned patterns of language, so given some words it predicts what words should come next — like very advanced autocomplete. It does not think like a human; it calculates probabilities. Our model, Qwen 2.5, has 3 billion parameters, which are like adjustable knobs the training tuned.

Q7. How does it know facts about your school/organisation? Did you train it?

Answer: We did not retrain the model — instead we use a technique called RAG, Retrieval-Augmented Generation. We fed our website pages and documents into a folder of text files. When a visitor asks something, the system finds the 3 most relevant pieces of text using embeddings — numbers that represent meaning — and gives them to the AI as 'FACTS'. The AI is instructed to answer only from those facts, so it acts like an open-book exam.

Tip: RAG is the concept judges are most likely to probe. Practise explaining 'open-book exam' analogy.

Q8. What are embeddings?

Answer: An embedding turns a sentence into a long list of numbers so that sentences with similar meaning get similar numbers. For example 'What time does school start?' and 'When do classes begin?' get numbers close to each other. We measure closeness with cosine similarity, which is like measuring the angle between two arrows.

Q9. How does the speech recognition work?

Answer: We use Whisper, an open-source speech-to-text model. It converts the sound wave into a spectrogram — a picture of the frequencies over time — and a neural network reads that picture and writes out the words. We use the multilingual 'small' version because it understands Indian-accented English better than the English-only version.

Q10. How does the face detection work? Does it know who I am?

Answer: The camera uses OpenCV with a Haar cascade — a classic pattern-matching method that looks for the light-and-dark pattern of eyes, nose, and mouth. That only detects that a face is present, so Vyoma knows to greet you. Optionally, if a visitor agrees to enroll, a face-

recognition library can store a face embedding so returning visitors are greeted by name. Enrolment is consent-based and stored only on this machine.

Q11. What is 'agentic'? You call it an agentic receptionist.

Answer: Agentic means the AI does not just chat — it can decide to take actions using tools we gave it. Vyoma has tools to log a visit, save a visitor's details, notify a staff member, and end the conversation. The model itself decides when to call each tool, like a real receptionist deciding when to pick up the phone.

Q12. Where is the data stored?

Answer: In a small database file called SQLite on the laptop itself. It stores visitors, their visits, and notifications. Nothing is uploaded anywhere.

Q13. What programming language did you use?

Answer: Python. The project has separate modules for speech-to-text, the AI agent, the knowledge base, text-to-speech, vision, and the database — about nine files that each do one job.

C. Offline, Privacy and Comparison Questions

Q14. How is this different from Alexa, Siri or ChatGPT?

Answer: Three big differences. First, those send your voice to company servers on the internet; Vyoma runs 100% on this laptop, so it works even with the Wi-Fi off. Second, they know general world knowledge; Vyoma is grounded in our own documents, so it answers about our organisation specifically. Third, Vyoma takes receptionist actions — logging visits and notifying staff — not just answering questions.

Tip: Offer to demonstrate: switch off Wi-Fi in front of the judge and ask Vyoma a question. Very convincing!

Q15. Why does offline matter?

Answer: Privacy — visitor names, faces and conversations never leave the building. Cost — no subscription or API charges. Reliability — it works during internet outages, and in rural areas with poor connectivity. This makes it practical for small schools, clinics and offices in India.

Q16. Is it safe to store people's faces and names? What about privacy laws?

Answer: We thought about this. Face enrolment is optional and only with the person's consent. The face is stored as an embedding — a list of numbers — not as a photo, and everything stays in a local database. In a real deployment we would also add a data-retention rule, like deleting visit records after some months, and follow India's DPDP data-protection act.

Tip: Judges score ethics awareness highly — do not skip this preparation.

Q17. Will this replace human receptionists and take away jobs?

Answer: We see it as a helper, not a replacement. It can handle repetitive questions and after-hours visitors, while humans handle emotions, judgement and unusual situations. It is like an ATM — banks still have staff, but routine work became faster.

D. Limitations and 'What If' Questions (judges test honesty here)

Q18. What happens if it mishears or gives a wrong answer?

Answer: It can happen. If the room is noisy, Whisper may transcribe wrongly, and then the answer will be off. We reduce mistakes by giving the speech recogniser a vocabulary hint with our organisation's names, and the AI is instructed to answer only from the retrieved facts. If it doesn't find relevant facts, it says it doesn't know instead of guessing.

Q19. Can the AI 'hallucinate' — make things up?

Answer: Yes, all language models can. That is exactly why we use RAG: the model must ground its answer in the FACTS we retrieve from our documents, and we keep its 'temperature' setting low so it stays factual rather than creative. It reduces hallucination a lot, but no system is perfect.

Q20. Does it work in noisy places like this exhibition hall?

Answer: Noise is our biggest challenge. We use an energy threshold so it ignores quiet background sound, and the visitor should stand close to the microphone. In a real lobby we would use a directional microphone. For today's demo we also have a keyboard text mode as backup.

Tip: ALWAYS have the --text mode ready at the fest. Demos fail in noisy halls; a backup saves your presentation.

Q21. Which languages does it support? Can it speak Malayalam/Hindi?

Answer: Right now it listens and replies in English, and Whisper understands Indian accents well. We already have a Hindi voice installed for the speaking part. Full Malayalam or Hindi conversation is possible because Whisper is multilingual — that is on our future plan.

Q22. How fast is it? Why is there a delay?

Answer: Each turn takes a few seconds: listening, converting speech to text, the AI thinking, and generating the voice. The AI model stays loaded in the graphics card memory all the time so we don't waste time reloading it. On a more powerful computer it would be faster.

Q23. What if two people talk at the same time, or someone interrupts it?

Answer: Currently it handles one speaker at a time and finishes its sentence before listening again. Handling interruptions — called barge-in — and separating two voices are known hard problems, and they are on our improvement list.

Tip: Naming the limitation with its technical term ('barge-in') impresses judges.

E. Hardware, Cost and Practicality

Q24. What hardware does it need? How much does it cost?

Answer: Just a laptop with a basic graphics card (4 GB), a webcam and a microphone — we validated it on a repurposed laptop. Since all the software is free and open source (Whisper, Qwen via Ollama, Piper, OpenCV, SQLite), the only cost is hardware many schools and offices already have. Cloud assistants cost a monthly fee forever; this costs nothing to run.

Q25. Why did you choose such a small AI model instead of a big one like GPT?

Answer: Big models need expensive servers or internet APIs. We chose Qwen 2.5 with 3 billion parameters, compressed to 4-bit, so it fits in a 4 GB graphics card. A receptionist doesn't need to write essays — it needs short, correct, polite answers, and a small model with RAG does that well. It's about choosing the right tool for the job.

Q26. How do you update its knowledge when something changes?

Answer: Very easily — we run one ingest command with a new PDF or the website address, or simply edit a text file. The system detects the change and rebuilds its memory automatically. No retraining is needed.

Q27. Where can this actually be used?

Answer: School and college receptions, hospital enquiry desks, government office help counters, libraries, museums, and small businesses — anywhere people ask the same questions repeatedly and privacy or internet cost matters.

F. About You (judges always ask students this)

Q28. What was your contribution? Did someone help you?

Answer: Be honest and specific. Example: 'We chose the idea, collected and cleaned the school information for the knowledge base, tested the voices and picked the best one, tuned the settings like silence detection, tested it with many questions, and prepared this demo. We took guidance from our mentor for setting up the AI models.' Judges can tell when students did not do the work — honesty plus understanding beats exaggeration.

Tip: Agree between yourselves beforehand on who did what, and each explain your own part.

Q29. What was the hardest part?

Answer: Pick a real one, for example: getting the microphone to know when the speaker has finished (silence detection), making the AI stop giving long essay answers instead of short receptionist replies, or making speech recognition work with our accents. Describe the problem and how you fixed it — that is what judges want to hear.

Q30. What did you learn from this project?

Answer: Examples: how speech recognition, language models and text-to-speech connect into a pipeline; what RAG and embeddings are; that AI can run on ordinary computers, not just the cloud; and that testing with real users reveals problems you never expected.

Q31. What will you improve next?

Answer: Our roadmap: an animated face on screen with lip-sync, so it feels friendlier; Malayalam and Hindi conversation; measuring and reducing the response delay; and a proper stand with a directional microphone for noisy lobbies.

G. Rapid-Fire One-Liners (memorise these facts)

If the judge asks...	Say...
Speech-to-text?	faster-whisper 'small' model, runs on the CPU
The AI brain?	Qwen 2.5, 3 billion parameters, 4-bit, runs locally via Ollama
Text-to-speech?	Piper, with US, UK and Hindi voices installed
Face detection?	OpenCV Haar cascade; optional face recognition with consent
Knowledge search?	nomic-embed-text embeddings + cosine similarity, top-3 facts
Database?	SQLite — visitors, visits, notifications
Internet needed?	No. 100% offline. We can switch off Wi-Fi and show you.
Language written in?	Python
Hardware?	One laptop: 4 GB GPU, webcam, microphone
Response time?	A few seconds per turn

H. Demo-Day Checklist

- Charge the laptop fully and carry the charger; ask organisers for a power socket.
- Test the microphone and speaker at the venue before judging starts.
- Keep text mode (keyboard chat) ready as a noisy-hall backup.

- Prepare 3 demo questions Vyoma answers well from the knowledge base — and practise them.
- One student explains while the other operates; then swap for the next judge.
- If Vyoma gives a wrong answer during the demo, don't panic — explain WHY it can happen (noise, retrieval miss). Turning a failure into an explanation earns respect.
- Bring a printed architecture diagram (mic → speech-to-text → AI + facts → text-to-speech → speaker).
- Offer the Wi-Fi-off demonstration to prove it is offline.